

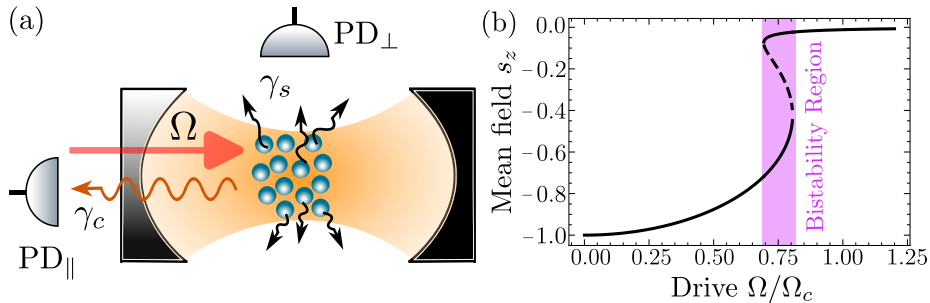
Proposal for Final Project: Predicting open system quantum dynamics by learning on quantum trajectories

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Physics (short): Driven-dissipative cavity QED



(a) System under study (b) Mean-field solution and bistability region

Main equation – Master equation

$$\dot{\rho} = -i[\mathcal{H}, \rho] + \frac{\gamma_c}{2} \mathcal{D}[\hat{S}_-]\rho + \frac{\gamma_s}{2} \sum_n \mathcal{D}[\hat{\sigma}_n]\rho, \quad \mathcal{H} = 2\Omega\hat{S}_x \quad (1)$$

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How to solve the master equation? Answer: MCWF method

Generally: Solve for density matrix: $4^N \times 4^N$ elements, N is the number of qubits.
Instead: Probabilistic unraveling of the master equation into quantum trajectories

$$|\psi(t + dt)\rangle = \begin{cases} \frac{\hat{S}_- |\psi(t)\rangle}{\sqrt{p_-}} & \text{with probability } p_- = \gamma_c dt \langle \psi(t) | \hat{S}_+ \hat{S}_- | \psi(t) \rangle \\ \frac{\hat{\sigma}_n |\psi(t)\rangle}{\sqrt{p_n}} & \text{with probability } p_n = \gamma_s dt \langle \psi(t) | \hat{\sigma}_n^\dagger \hat{\sigma}_n | \psi(t) \rangle \\ \frac{e^{-i\mathcal{H}_{NH}dt} |\psi(t)\rangle}{\sqrt{p_0}} & \text{with probability } p_0 = 1 - p_- - \sum_n p_n \end{cases} \quad (2)$$

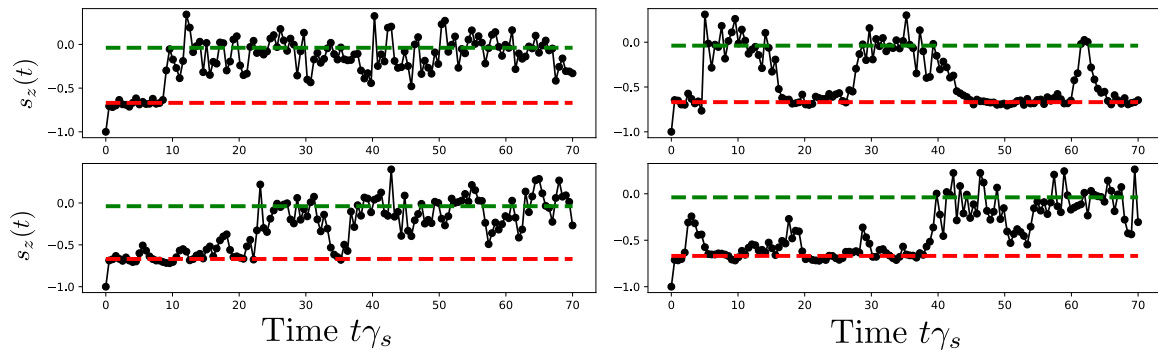
✓ Saves memory: $2^N \times 2^N$ elements

✗ Loses Time: Need to average over many trajectories to get the density matrix.

In our work we found that a bistability is seen in a single trajectory, but the bistability is lost in the density matrix average.

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How typical trajectory looks like (in bistability case)

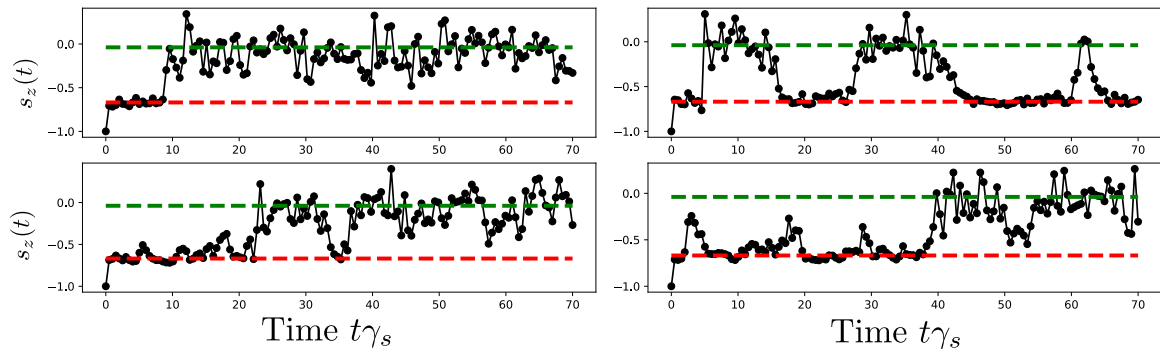


4 examples of quantum trajectories $N = 18$ (best I can get), $\Omega = 0.73\Omega_c$ (bistability region), $\gamma_c = 10\gamma_s$. Red and Green lines are the mean-field solutions. Generated with QuTiP (python library for quantum simulations).

Note: average of $N_{tr} \approx 100$ trajectories gives the density matrix solution, which is a smooth curve (no bistability).

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Problem statement



4 examples of quantum trajectories $N = 18$ (best I can get), $\Omega = 0.73\Omega_c$ (bistability region), $\gamma_c = 10\gamma_s$.

Goal of the project

Create a model to generate trajectories similar to quantum trajectories generated by Monte Carlo wave-function simulations in QuTiP

Why this matters?

